

L1.2 Limits

3 · A general flowchart for evaluating limits

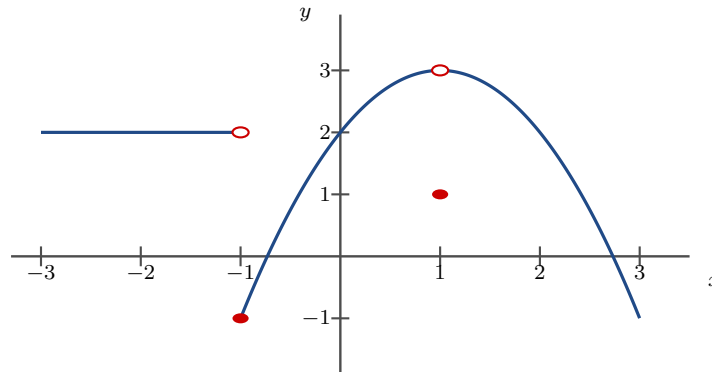
1. Open **Limit Diagnostic in Quarto**. Try at least 6 flash cards **six**: at least **two** that return $\frac{0}{0}$, one that returns a number, one that returns $\frac{b}{0}$, and one from **below the divider**.

Flash Card	$f(a)$ returned	which line of the chart	the limit

2. One flash card returns a perfectly clean number and the limit still does not exist. Find it. What is true about that function near a that is not true of the other clean-number cards?
3. Work on **both** oscillation flash cards. They have the same zeros and oscillate in a similar manner, but only one of them has a limit. How can you determine what the correct answer is?

1 · What a limit is

4. From the graph:



- (a) $\lim_{x \rightarrow -1^-} f(x)$ (b) $\lim_{x \rightarrow -1^+} f(x)$ (c) $\lim_{x \rightarrow -1} f(x)$ (d) $f(-1)$ (e) $\lim_{x \rightarrow 1} f(x)$ (f) $f(1)$
(g) One of the two marked x -values gets the shorthand H @. Which one?

2 · Diagnosing limits

5. Run the diagnostic on each. Write what came back and what the chart says to do next — **do not evaluate yet**. Then finish the two that need no rewriting at all.

(a) $\lim_{x \rightarrow 4} \frac{x^2 - 3x - 4}{x - 4}$ (b) $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x + 2}$ (c) $\lim_{x \rightarrow -2} \frac{x}{x + 2}$ (d) $\lim_{x \rightarrow 0} \sqrt{\frac{4x + 9}{x + 1}}$

6. Factor and cancel.

Stop. Before you cancel — what does a cancellation leave behind on the graph, and what is the limit equal to?

$$(a) \lim_{x \rightarrow 4} \frac{x^2 - 3x - 4}{x - 4} \qquad (b) \lim_{x \rightarrow 3} \frac{2x^2 - 5x - 3}{x^2 - 9}$$

For (b), also give the full shorthand for the graph: HA @, Z@, H @, VA @.

7. Square-rooty, so rationalize.

$$\lim_{h \rightarrow 0} \frac{\sqrt{4+h} - 2}{h}$$

8. Both of these return $\frac{b}{0}$. For each: LHL, then RHL, then the two-sided answer.

Stop. Before you write ∞ — what does $\frac{b}{0}$ tell you, and what does it not tell you?

$$(a) \lim_{x \rightarrow -2} \frac{x}{x+2} \qquad (b) \lim_{x \rightarrow 1} \frac{5}{(x-1)^2}$$

9. You are told only that

$$4x - 3 \leq f(x) \leq x^2 + 1 \quad \text{for every } x.$$

Find $\lim_{x \rightarrow 2} f(x)$, and say why you never needed a formula for f .

4 · The laws

10. Let

$$f(x) = 3x - 1 \quad (x \neq 1) \qquad g(x) = \begin{cases} (x - 1)^2 & x \neq 2 \\ 0 & x = 2 \end{cases}$$

Evaluate:

$$(a) \lim_{x \rightarrow 2} [4f(x) - g(x)] \qquad (b) \lim_{x \rightarrow 1} \sqrt{3f(x)} \qquad (c) \lim_{x \rightarrow 0} \frac{f(x)}{g(x)}$$

11. Same f and g : $\lim_{x \rightarrow 1} \frac{f(x)}{g(x)}$. The quotient law does **not** apply here — which hypothesis fails? Then go back to the function itself and work both sides.